

ELSAYED GHONAIM

MACHINE LEARNING ENGINEER | GENERATIVE AI & PRODUCTION ML

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3 years of hands-on experience building and deploying generative AI, natural language processing, computer vision, and predictive modeling systems. Develops production-oriented Python services with PyTorch, TensorFlow, FastAPI, Flask, Docker, Kubernetes, and CI/CD, with public projects in LLM serving and multimodal document extraction.

TECHNICAL SKILLS

Programming: Python, C++, Bash, Perl

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, CNN, ResNet50, YOLO, LSTM, Feature Engineering, Model Evaluation, Time Series Forecasting

Generative AI & NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Transformers, Hugging Face, vLLM, llama.cpp, Fine-Tuning, LoRA, OCR, Vision-Language Models (VLMs)

MLOps & Deployment: FastAPI, Flask, REST APIs, Model Serving, Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Pytest, Linux, Git

Data, Vision & Geospatial: Pandas, NumPy, Statsmodels, OpenCV, Rasterio, Data Pipelines, Remote Sensing, Satellite Imagery, Geospatial Analysis

PROFESSIONAL EXPERIENCE

AI Engineer | Z2Data | Egypt | Sep 2025 - Present

- Architected a Retrieval-Augmented Generation (RAG) system that integrated proprietary datasets into an internal chatbot, improving response grounding and reducing hallucinations.
- Fine-tuned transformer-based Large Language Models (LLMs) on domain-specific datasets and partnered with product teams to improve contextual relevance for enterprise use cases.
- Built OCR and document-processing pipelines that converted scanned, unstructured content into structured, retrieval-ready data.
- Collaborated with backend engineers to productionize AI/ML capabilities through REST APIs and internal enterprise services.

Machine Learning Engineer | Arab Company for Training and Artificial Intelligence Applications (ACTAIA) | Cairo, Egypt | Jul 2025 - Dec 2025

- Developed end-to-end AI applications spanning machine learning model training, backend services, and frontend integration.
- Designed and deployed a Smart Fire Alert System and a Multi-Task Medical Image Classifier, supporting delivery from model development through application integration.

Co-Founder and AI/ML Team Lead | Larvest | Cairo, Egypt | Jan 2024 - Jul 2025

- Led a cross-functional team of five engineers and remote sensing specialists, setting technical direction for predictive machine learning systems in precision agriculture.
- Designed a YOLO-based computer vision pipeline for crop-disease detection from drone imagery and applied model optimization and data-processing techniques.
- Deployed machine learning models through Flask REST APIs and supported integration with web dashboards for end-user access to AI insights.

PROJECTS

ForgeAI - Dual-Backend LLM Serving Platform

Python, FastAPI, vLLM, llama.cpp, Docker, Kubernetes, Prometheus, GitHub Actions, Pytest | github.com/elsayedghoonaim/ForgeAI

- Engineered a Python CLI and FastAPI service that unified two Large Language Model inference backends - vLLM and llama.cpp - with automatic model-format selection and OpenAI-compatible endpoints.
- Implemented production-oriented model-serving infrastructure with Docker, Kubernetes, Prometheus metrics, authentication, rate limiting, health/readiness probes, GitHub Actions CI, and Pytest coverage.

Universal PDF-to-Markdown Extraction Pipeline

Python, OCR, Vision-Language Models, OpenAI, Google Gemini, Ollama, vLLM, PyMuPDF | github.com/elsayedghoonaim/ParsingPDFs

- Built a Python document-AI pipeline that classifies native-text and scanned PDFs, routes them through text extraction or Vision-Language Model (VLM) OCR, and produces structured Markdown.
- Designed a provider abstraction supporting five model integrations - Google, OpenAI, Ollama, vLLM, and custom endpoints - with configurable concurrency, cost estimation, state tracking, and unit tests.

AI-Enhanced Satellite Tracking System - Graduation Project

PyTorch, Random Forest, Flask, SGP4, TLE Data, Orbital Mechanics

- Built an error-correction machine learning pipeline around the SGP4 orbit-propagation model using orbital mechanics, satellite properties, TLE data, and space-weather indices.
- Developed and evaluated Random Forest regression and a PyTorch Deep Neural Network, then exposed on-demand orbit corrections through a Flask API.

Multi-Task Medical Image Classifier

PyTorch, CNN, ResNet50, Computer Vision

- Built a multi-headed ResNet50 CNN for three medical-imaging tasks and achieved 98.34% test accuracy through custom data augmentation and masked-loss optimization.

EDUCATION

Bachelor Degree in Navigation Science and Space Technology | Beni-Suef University | New Beni-Suef City, Egypt | Sep 2020 - Jul 2025

GPA: 3.0 | Relevant Coursework: Python, C++, Orbital Mechanics, Remote Sensing, Flight Dynamics, Optimal Trajectories, Space Guidance and Control Systems

CERTIFICATIONS

Generative AI with Large Language Models - AWS, 2024

Deep Learning Specialization - DeepLearning.AI, 2024

TensorFlow Developer Professional Certificate - DeepLearning.AI, 2024

Machine Learning Specialization - DeepLearning.AI, 2023

Additional Credentials

Global Nominee - NASA Space Apps Challenge, 2023

Machine Learning Workshop Instructor - SPIN Organization, 2024